

ReadMe File

Title: The Dire Effects of the Lack of Monetary and Fiscal Coordination

Journal of Monetary Economics

Authors: Francesco Bianchi and Leonardo Melosi

Date: 03/11/2021

The file main.m produces Figures 3, 4, 6, 7, 8, and 9 of the paper. It takes the following variables as **inputs**:

- a) Lines 21-24: set the initial stock of debt (**b0**), the length of the IRF simulations (**nirf**), the duration of the ZLB/low demand period in the simulations (**xx**), the duration of the conflict period (**ft**).
- b) Lines 25-37 set the graphical specifications for the figures.
- c) Line 39: Set **cond_moments=1**, if you want the code to compute and save the conditional moments needed for calculating macro volatilities and welfare (time consuming!). If set equal to zero, the code takes the conditional moments from the subfolder **\CondMoms**.
- d) Lines 45-58 upload the parameters of the model which can be found in different tabs of the excel file **\Params\Params.xlsx**. Note that the variable **p_st** is a matrix with as many as rows the parameters used in the file containing the linearized equations of the model, **\Model\gammas2771.m**. The ten columns of this matrix contain the model parameters conditional on each of the ten regimes (We list the ten regimes at the end of the document.) The parameter vectors **p_ss**, **p_er**, and **o_er** are defined in **\Model\gammas2771.m** and do not depend on regimes. The first four elements of the vector **oparams** are used to

construct the transition matrix of the ten regimes in `\Model\DoConflictMatrix.m`. The first element is the probability of staying out of the ZLB/low demand regime. The first element is the probability of staying in the ZLB regime. The third element is the persistence of the three conflict regimes (regimes 3-5). The fourth element is the probability that one of the three conflict regimes (regime 5) will be ended with a switch to the monetary-led Regime (regime 1). The fifth element controls the size of the demand shock that triggers the ZLB Regime (Regimes 6-10). The sixth element sets the lower value of the discount factor beta for the exercise shown in Figure 7 of the paper. The eight element is the parameter that pins down the markup at steady state, which is needed to compute social welfare.

The matrix `HB` is the transition matrix for the three regimes estimated in Bianchi and Melosi (2017 AER) (Regime 1: Monetary Led, Regime 2 Fiscally Led, and Regime 3 ZLB Regime). The vector `msel` contains model options. The more relevant are the sixth element determining which model to select (in this case 2771 -> `\Model\gammas2771.m`). The thirteen element of `msel` picks the solution method (in this case Farmer, Waggoner, and Zha or gensys) and check MSS. The fifteen element of `msel` sets whether the code has to build the variance and covariance matrix and the mapping from the model to the observables.

`MATlc{7}` sets the initial conditions for the solution algorithm. If it is an empty cell, the code initialize the solution algorithm automatically using gensys by Chris Sims.

LIST OF REGIMES

Regime 1: Monetary-led regime.

Regime 2: Fiscally-led regime.

Regime 3: Conflict regime that will end with a switch to the Monetary-led regime (Regime 1).

Regime 4: Conflict regime that will end with a switch to the Fiscally-led regime (Regime 2).

Regime 5: Conflict regime that will end with uncertain resolution - *not used in the published version of the paper.*

Regime 6: ZLB Regime with exit to the Monetary-led regime (Regime 1).

Regime 7: ZLB Regime with exit to the Fiscally-led regime (Regime 2).

Regime 8: ZLB Regime with exit to the Conflict regime and then Monetary-led regime (Regime 3).

Regime 9: ZLB Regime with exit to the Conflict regime and then Fiscally-led regime (Regime 4).

Regime 10: ZLB Regime with exit to the Conflict regime with uncertain resolution (Regime 5) - *not used in the published version of the paper.*